

NTC_CC105 - Test Case Documentation

Smart Campus Student Resource Management System | BSIT 2.4 | Instructor: Darwin Vargas

Executed Test Cases

No.	Feature	Input	Expected Output	Actual Output	Status
1	Add Student - Normal Input	Valid record: ID 1001	Record added successfully.	1001 added and displayed.	PASS
2	Duplicate Record	Add ID 1001 again	Duplicate ID is rejected.	Duplicate ID rejected.	PASS
3	Display Students	Display after five records	All stored records are listed.	Five records displayed; Total Students: 5.	PASS
4	Search Success	Student ID 1003	Matching record is displayed.	Record for 1003 displayed.	PASS
5	Search Failure	Student ID 9999	Record not found message.	Record not found.	PASS
6	Update Student	ID 1002; Year Level = 2nd Year	Record is updated.	1002 updated successfully.	PASS
7	Delete Student	Delete ID 1004	Record removed; remaining records shift.	1004 removed; four remained.	PASS
8	Invalid Numerical Input	abc as menu input	Invalid input is rejected.	Invalid input rejected.	PASS
9	Invalid Resource ID	R-99	Invalid Resource ID is rejected.	Resource not found.	PASS
10	Empty Required Field	ID 2001; blank Name	Empty required field is rejected.	Name cannot be empty.	PASS
11	Maximum Capacity	Temporary MAX_STUDENTS = 2; third add attempt	Program rejects additional record.	Maximum student capacity reached.	PASS
12	Matrix Processing	Display/analyze 4 x 4 matrix	Correct row/column totals are shown.	Rows 28,25,23,21; columns 30,19,28,20; total 97.	PASS
13	Dynamic Memory Cleanup	Exit program	Allocated resource array is released.	Dynamic memory released successfully.	PASS

Note: For the maximum-capacity validation, the limit was temporarily changed to 2 only for execution testing. The final source code was restored to MAX_STUDENTS = 50 and recompiled.